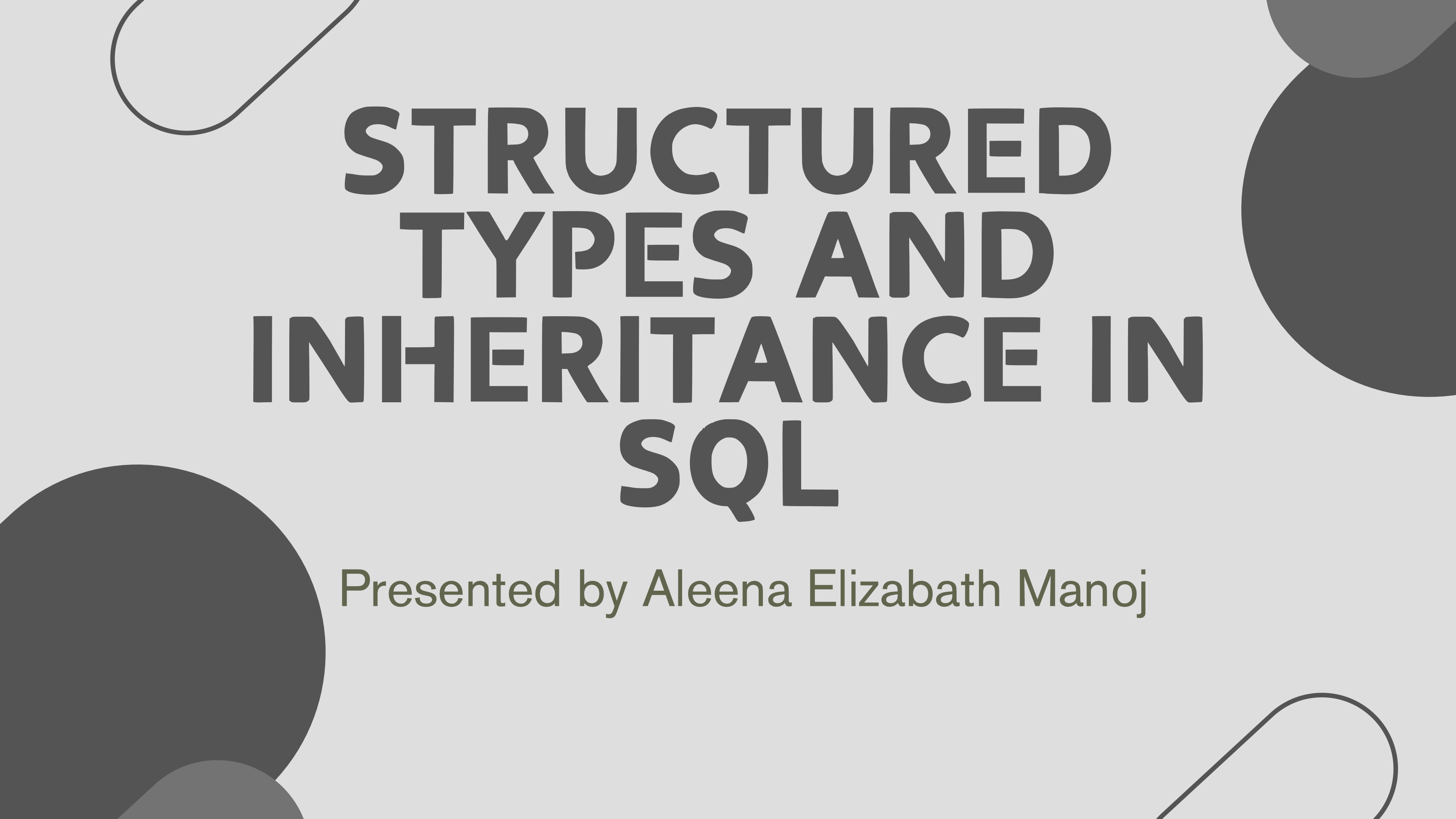




STRUCTURED TYPES AND INHERITANCE IN SQL

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INTRODUCTION

- A structure type in SQL allows users to create their own data types, combining multiple attributes into a single compound data type.
- A structured type can be used as the type of a table, view, or column.
- They are also called User defined types.

```
create type Name as  
  (firstname      varchar(20),  
   lastname      varchar(20))  
final
```

```
create type Address as  
  (street        varchar(20),  
   city          varchar(20),  
   zipcode       varchar(20))  
not final
```

- Note: **final** and **not final** indicate whether subtypes can be created

We can now use these types to create composite attributes in a relation, by simply declaring an attribute to be of one of these types. For example, we could create a table person as follows

Structured types can be used to create tables with composite attributes

```
create table customer (  
  name      Name,  
  address   Address,  
  dateOfBirth date)
```

Dot notation used to reference components: *name.firstname*

- We can also create a table whose rows are of a user-defined type. For example, we could define a type `PersonType` and create the table `person` as follows:

```
create type PersonType as (  
    name Name,  
    address Address,  
    dateOfBirth date)  
    not final  
create table person of PersonType;
```

- A structured type can have methods defined on it. We declare methods as part of the type definition of a structured type

```
create type PersonType as (  
    name Name,  
    address Address,  
    dateOfBirth date)  
not final  
method ageOnDate(onDate date)  
    returns interval year;
```

We create the method body separately:

```
create instance method ageOnDate (onDate date)  
    returns interval year  
    for PersonType  
begin  
    return onDate – self.dateOfBirth;  
end
```


- for clause indicates which type this method is for, while the keyword instance indicates that this method executes on an instance of the Person type.
- The variable self refers to the Person instance on which the method is invoked.
- constructor functions are used to create values of structured types.
- A function with the same name as a structured type is a constructor function for the structured type

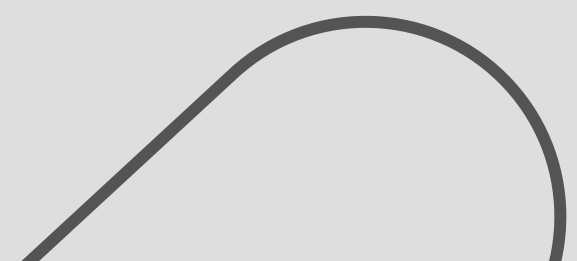
We can also create tuples using the user defined types:

```
insert into Person
  values
    (new Name('John', 'Smith'),
     new Address('20 Main St', 'New York', '11001'),
     date '1960-8-22');
```

INHERITANCE IN SQL

Allows a new object type (subtype) to inherit attributes and methods from an existing object type (supertype).

- Subtypes inherit attributes and methods from the supertype.
- A subtype can add new attributes or methods.
- A subtype can override inherited methods

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- We may want to store extra information in the database about people who are students, and about people who are teachers.
 - Since students and teachers are also people, we can use inheritance to define the student and teacher types in SQL
 - Both Student and Teacher inherit the attributes of Person —namely, name and address.
 - Student and Teacher are said to be subtypes of Person, and Person is a supertype of Student, as well as of Teacher.

```
create type Person  
  (name varchar(20),  
   address varchar(20));
```

```
create type Student  
  under Person  
  (degree varchar(20),  
   department varchar(20));
```

```
create type Teacher  
  under Person  
  (salary integer,  
   department varchar(20));
```

Multiple inheritance:

```
create type TeachingAssistant  
under Student, Teacher;
```

- The SQL standard requires an extra field at the end of the type definition, whose value is either final or not final.
- The keyword final says that subtypes may not be created from the given type, while not final says that subtypes may be created

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THANK YOU!!